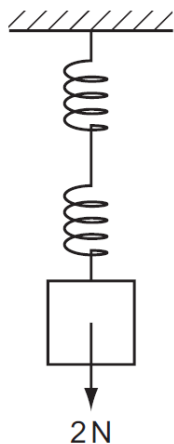


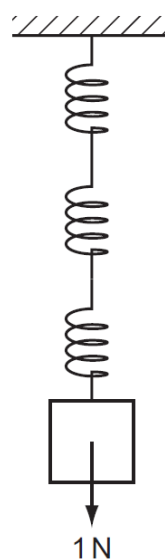
- 5** Several identical springs, each having the same spring constant, are joined in four arrangements. A different load is applied to each arrangement.

Which arrangement has the largest extension?

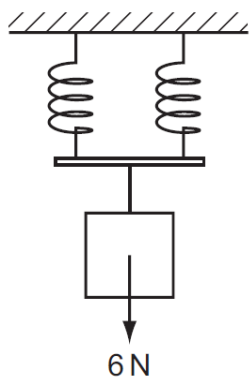
A



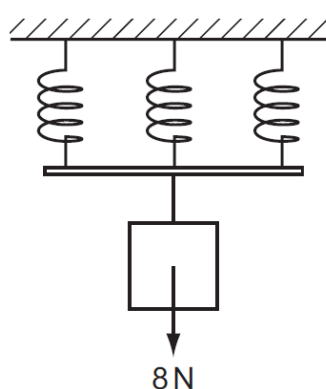
B



C



D



Answer: A

Let k be the spring constant of each of the identical springs.

For a spring system, $F_{\text{net}} = k_{\text{eff}} (e)$

where F_{net} is the load

k_{eff} is the effective spring constant

e is the extension of the arrangement

For A: $e = 2 / (0.5k) = 4 / k$ (largest)

For B: $e = 1 / (k / 3) = 3 / k$

For C: $e = 6 / 2k = 3 / k$

For D: $e = 8 / 3k = 2.7 / k$ (smallest)

